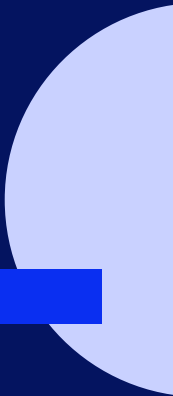
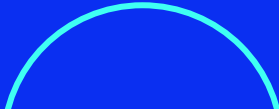


Introducción a MongoDB

Módulo 5



Configuración del conjunto de réplicas

Configurar conjunto de réplicas

Preparación ambiental

Las IP de cada nodo en el conjunto de réplica son las siguientes:

localhost:27017

localhost:27018

localhost:27019

Se van a crear tres carpetas para alojar las tres instancias de servidor mongoDB.

```
mkdir \data\rs1 \data\rs2 \data\rs3
```

Luego, se inician las instancias de servidor de base de datos en tres consolas distintas:

```
mongod --replSet m101 --dbpath \data\rs1 --port 27017  
mongod --replSet m101 --dbpath \data\rs2 --port 27018  
mongod --replSet m101 --dbpath \data\rs3 --port 27019
```



Se mostrará la operación en una consola con múltiples ventanas, en donde se van a abrir 3 layers con la inicialización del servidor **mongod** y su cliente respectivo **mongo**. En este punto, las tres instancias de MongoDB se han iniciado como conjuntos de réplica, pero aún no se comunicarán entre sí, y aún se requiere alguna configuración.



```
C:\>
λ mongod --replSet m101 --dbpath \data\rs1 --port 27017
{"t":{"$date":"2022-05-21T09:28:08.340-03:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"-", "msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-21T09:28:08.862-03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"main", "msg":"Initialized wire specification"}

C:\cmdr_mini>
λ mongo --port 27017
MongoDB shell version v5.0.8
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("3db4519d-d679-4b41-8165-c1a5b1762462") }
MongoDB server version: 5.0.8

C:\>
λ mongod --replSet m101 --dbpath \data\rs2 --port 27018
{"t":{"$date":"2022-05-21T09:29:38.631-03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"-", "msg":"Initialized wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}

C:\cmdr_mini>
λ mongo --port 27018
MongoDB shell version v5.0.8
connecting to: mongodb://127.0.0.1:27018/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("8f219361-a9cc-4b5c-b03b-303ea7e64407") }
MongoDB server version: 5.0.8

C:\>
λ mongod --replSet m101 --dbpath \data\rs3 --port 27019
{"t":{"$date":"2022-05-21T09:31:39.133-03:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"main", "msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-21T09:31:39.655-03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"main", "msg":"Initialized wire specification"}

C:\cmdr_mini>
λ mongo --port 27019
MongoDB shell version v5.0.8
connecting to: mongodb://127.0.0.1:27019/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("e66f8a6c-44ac-4fda-80e5-ac68adc54071") }
MongoDB server version: 5.0.8
```

Réplica de inicialización del conjunto

Sobre el shell de la primera instancia de MongoDB (port 27017), se ejecuta el método **rs.initiate()** para inicializar el conjunto de réplicas. Esta función necesita recibir una variable que establece la configuración y puesta en marcha del conjunto de réplica.

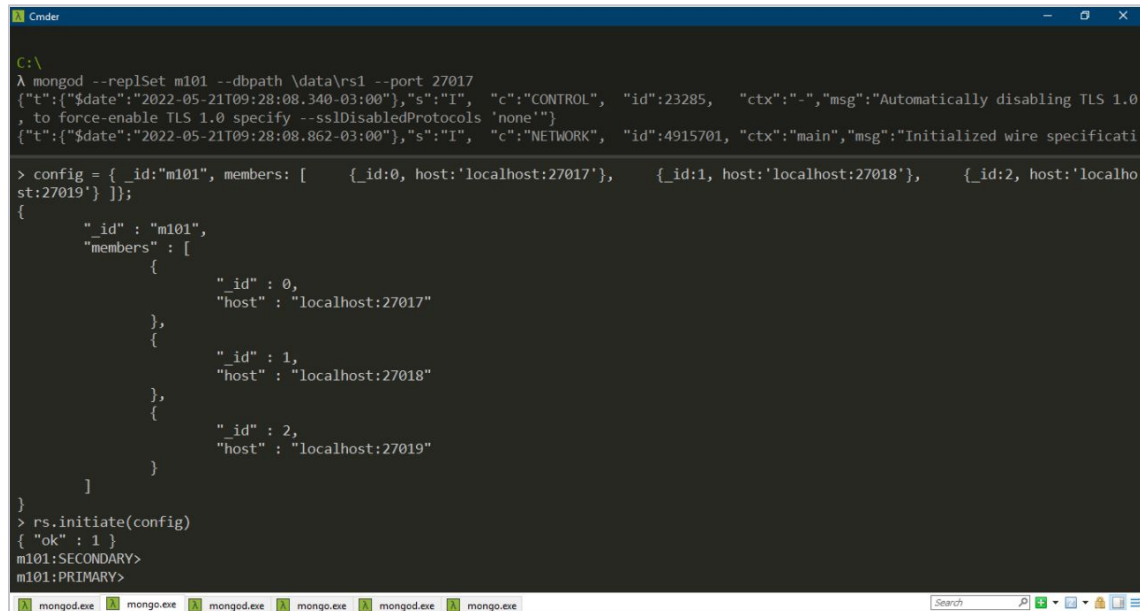
Primero se crea la variable en el *shell*

```
config = { _id:"m101", members: [
  { _id:0, host:'localhost:27017'},
  { _id:1, host:'localhost:27018'},
  { _id:2, host:'localhost:27019'}
  ]};
```

Luego se ejecuta la inicialización

```
rs.initiate(config)
```

Se tendrá:



```
C:\>
λ mongod --replSet m101 --dbpath \data\rs1 --port 27017
{"t":{"$date":"2022-05-21T09:28:08.340-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-", "msg":"Automatically disabling TLS 1.0
, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-21T09:28:08.862-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main", "msg":"Initialized wire specificati

> config = { _id:"m101", members: [    { _id:0, host:'localhost:27017'},    { _id:1, host:'localhost:27018'},    { _id:2, host:'localhost:27019'}  ] };
{
  "_id" : "m101",
  "members" : [
    {
      "_id" : 0,
      "host" : "localhost:27017"
    },
    {
      "_id" : 1,
      "host" : "localhost:27018"
    },
    {
      "_id" : 2,
      "host" : "localhost:27019"
    }
  ]
}
> rs.initiate(config)
{ "ok" : 1 }
m101:SECONDARY>
m101:PRIMARY>
```

Al principio el nodo inicia como secundario y, al cabo de unos segundos, se convertirá en el nodo primario. Los demás nodos (27018 y 27019) quedarán como secundarios.

```
C:\cmdr_mini
λ mongod --replSet m101 --dbpath \data\rs2 --port 27018
{"t":{"$date":"2022-05-21T09:29:38.631-03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"","msg":"Initialized wire specification",
"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
m101:SECONDARY> |
```

```
C:\cmdr_mini
λ mongod --replSet m101 --dbpath \data\rs3 --port 27019
{"t":{"$date":"2022-05-21T09:31:39.133-03:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"main","msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-21T09:31:39.655-03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"main","msg":"Initialized wire specification"}
m101:SECONDARY> |
```

Con los comandos **rs.status()** y **rs.conf()** sobre el cliente primario se verifica la configuración del conjunto de réplica.

rs.status

```
m101:PRIMARY> rs.status()
{
  "set" : "m101",
  "date" : ISODate("2022-05-21T12:47:27.648Z"),
  "myState" : 1,
  "term" : NumberLong(1),
  "syncSourceHost" : "",
  "syncSourceId" : -1,
  "heartbeatIntervalMillis" : NumberLong(2000),
  "majorityVoteCount" : 2,
  "writeMajorityCount" : 2,
  "votingMembersCount" : 3,
  "writableVotingMembersCount" : 3,
  "optimes" : {
    "lastCommittedOpTime" : {
      "ts" : Timestamp(1653137246, 1),
      "t" : NumberLong(1)
    },
  },
}
```


...

```
"lastCommittedWallTime" : ISODate("2022-05-21T12:47:26.843Z"),
"readConcernMajorityOpTime" : {
  "ts" : Timestamp(1653137246, 1),
  "t" : NumberLong(1)
},
"appliedOpTime" : {
  "ts" : Timestamp(1653137246, 1),
  "t" : NumberLong(1)
},
"durableOpTime" : {
  "ts" : Timestamp(1653137246, 1),
  "t" : NumberLong(1)
},
"lastAppliedWallTime" : ISODate("2022-05-21T12:47:26.843Z"),
"lastDurableWallTime" : ISODate("2022-05-21T12:47:26.843Z")
},
"lastStableRecoveryTimestamp" : Timestamp(1653137200, 1),
```

...

...

```
"electionCandidateMetrics" : {
  "lastElectionReason" : "electionTimeout",
  "lastElectionDate" : ISODate("2022-05-21T12:42:56.770Z"),
  "electionTerm" : NumberLong(1),
  "lastCommittedOpTimeAtElection" : {
    "ts" : Timestamp(1653136966, 1),
    "t" : NumberLong(-1)
  },
  "lastSeenOpTimeAtElection" : {
    "ts" : Timestamp(1653136966, 1),
    "t" : NumberLong(-1)
  },
  "numVotesNeeded" : 2,
  "priorityAtElection" : 1,
  "electionTimeoutMillis" : NumberLong(10000),
  "numCatchUpOps" : NumberLong(0),
  "newTermStartDate" : ISODate("2022-05-21T12:42:56.812Z"),
  "wMajorityWriteAvailabilityDate" : ISODate("2022-05-21T12:42:58.248Z")
},
```

...

...

```
"members" : [  
  {  
    "_id" : 0,  
    "name" : "localhost:27017",  
    "health" : 1,  
    "state" : 1,  
    "stateStr" : "PRIMARY",  
    "uptime" : 1159,  
    "optime" : {  
      "ts" : Timestamp(1653137246, 1),  
      "t" : NumberLong(1)  
    },  
    "optimeDate" : ISODate("2022-05-21T12:47:26Z"),  
    "lastAppliedWallTime" : ISODate("2022-05-21T12:47:26.843Z"),  
    "lastDurableWallTime" : ISODate("2022-05-21T12:47:26.843Z"),  
    "syncSourceHost" : "",  
    "syncSourceId" : -1,  
    "infoMessage" : "",  
    "electionTime" : Timestamp(1653136976, 1),
```

...

...

```
    "electionDate" : ISODate("2022-05-21T12:42:56Z"),
    "configVersion" : 1,
    "configTerm" : 1,
    "self" : true,
    "lastHeartbeatMessage" : ""
  },
  {
    "_id" : 1,
    "name" : "localhost:27018",
    "health" : 1,
    "state" : 2,
    "stateStr" : "SECONDARY",
    "uptime" : 281,
    "optime" : {
      "ts" : Timestamp(1653137246, 1),
      "t" : NumberLong(1)
    },
  },
```

...

...

```
    "optimeDurable" : {
      "ts" : Timestamp(1653137246, 1),
      "t"  : NumberLong(1)
    },
    "optimeDate" : ISODate("2022-05-21T12:47:26Z"),
    "optimeDurableDate" : ISODate("2022-05-21T12:47:26Z"),
    "lastAppliedWallTime" : ISODate("2022-05-21T12:47:26.843Z"),
    "lastDurableWallTime" : ISODate("2022-05-21T12:47:26.843Z"),
    "lastHeartbeat" : ISODate("2022-05-21T12:47:26.890Z"),
    "lastHeartbeatRecv" : ISODate("2022-05-21T12:47:26.390Z"),
    "pingMs" : NumberLong(0),
    "lastHeartbeatMessage" : "",
    "syncSourceHost" : "localhost:27017",
    "syncSourceId" : 0,
    "infoMessage" : "",
    "configVersion" : 1,
    "configTerm" : 1
  },
```

...

...

```
{
  "_id" : 2,
  "name" : "localhost:27019",
  "health" : 1,
  "state" : 2,
  "stateStr" : "SECONDARY",
  "uptime" : 281,
  "optime" : {
    "ts" : Timestamp(1653137246, 1),
    "t" : NumberLong(1)
  },
  "optimeDurable" : {
    "ts" : Timestamp(1653137246, 1),
    "t" : NumberLong(1)
  },
  "optimeDate" : ISODate("2022-05-21T12:47:26Z"),
  "optimeDurableDate" : ISODate("2022-05-21T12:47:26Z"),
  "lastAppliedWallTime" : ISODate("2022-05-21T12:47:26.843Z"),
  "lastDurableWallTime" : ISODate("2022-05-21T12:47:26.843Z"),
```

...

...

```
        "lastHeartbeat" : ISODate("2022-05-21T12:47:26.890Z"),
        "lastHeartbeatRecv" : ISODate("2022-05-21T12:47:26.390Z"),
        "pingMs" : NumberLong(0),
        "lastHeartbeatMessage" : "",
        "syncSourceHost" : "localhost:27017",
        "syncSourceId" : 0,
        "infoMessage" : "",
        "configVersion" : 1,
        "configTerm" : 1
    },
    "ok" : 1,
```

...

...

```
"$clusterTime" : {  
  "clusterTime" : Timestamp(1653137246, 1),  
  "signature" : {  
    "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAAA="),  
    "keyId" : NumberLong(0)  
  }  
},  
"operationTime" : Timestamp(1653137246, 1)  
}
```



rs.conf

```
m101:PRIMARY> rs.conf()
{
  "_id" : "m101",
  "version" : 1,
  "term" : 1,
  "members" : [
    {
      "_id" : 0,
      "host" : "localhost:27017",
      "arbiterOnly" : false,
      "buildIndexes" : true,
      "hidden" : false,
      "priority" : 1,
      "tags" : {
      },
      "secondaryDelaySecs" : NumberLong(0),
      "votes" : 1
    },
  ],
}
```

```
{
  "_id" : 1,
  "host" : "localhost:27018",
  "arbiterOnly" : false,
  "buildIndexes" : true,
  "hidden" : false,
  "priority" : 1,
  "tags" : {
  },
  "secondaryDelaySecs" : NumberLong(0),
  "votes" : 1
},
```

```
{
  "_id" : 2,
  "host" : "localhost:27019",
  "arbiterOnly" : false,
  "buildIndexes" : true,
  "hidden" : false,
  "priority" : 1,
  "tags" : {
  },
  "secondaryDelaySecs" : NumberLong(0),
  "votes" : 1
},
```

...

```
"protocolVersion" : NumberLong(1),
"writeConcernMajorityJournalDefault" : true,
"settings" : {
  "chainingAllowed" : true,
  "heartbeatIntervalMillis" : 2000,
  "heartbeatTimeoutSecs" : 10,
  "electionTimeoutMillis" : 10000,
  "catchUpTimeoutMillis" : -1,
  "catchUpTakeoverDelayMillis" : 30000,
  "getLastErrorModes" : {

  },
  "getLastErrorDefaults" : {
    "w" : 1,
    "wtimeout" : 0
  },
  "replicaSetId" : ObjectId("6288de46898db8af955e6b95")
}
```

**¡Sigamos
trabajando!**